1. A and B are sets. Prove $A \subseteq B$.

think: $A \subseteq B \iff \forall a \in A,\ a \in B$

$\iff$ If $a \in A$, then $a \in B$. $\iff$ If $a \notin B$, then $a \notin A$

Prop: $A \subseteq B$

Pf: Let $a \in A$.

$\vdots$

Thus, $a \in B$.

So $a \in A$ implies $a \in B$.

Thus, $A \subseteq B$.

Prop: $A \subseteq B$

Pf (by contrapos)

Let $x \notin B$.

$\vdots$

Thus, $x \notin A$.

So $x \notin B$ implies $x \notin A$. Thus, $A \subseteq B$.

Ex For every pair of sets A, B, prove $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$

Pf: Sppse A, B sets. Let $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$. So, by definition $X \in \mathcal{P}(A)$ or $X \in \mathcal{P}(B)$.

Case 1: $X \in \mathcal{P}(A)$. Thus, $X \subseteq A$. Thus, $X \subseteq A \cup B$. So $X \in \mathcal{P}(A \cup B)$.

Case 2: $X \in \mathcal{P}(B)$. . . . . (same) . . .

We showed $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$ implies $X \in \mathcal{P}(A \cup B)$. So $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

2. A is a set. Prove $a \in A$.

$$\mathcal{P}(A \cup B) = \{X : X \subseteq A \cup B\}$$

Used: $X \subseteq A \Rightarrow X \subseteq A \cup B$.

To prove $a \in \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \equiv y \bmod 6\}$

① show $a \in \mathbb{Z} \times \mathbb{Z}$.

So $a = (c_1, c_2)$, $c_1, c_2 \in \mathbb{Z}$.

② $c_1 \equiv c_2 \bmod 6$

To prove $a \in \{7n^2 + 1 : n \in \mathbb{Z}\}$

Find $n \in \mathbb{Z}$ so that $a = 7n^2 + 1$

3. Suppose A, B, and C are sets. Prove that if $B \subseteq C$, then $A \times B \subseteq A \times C$.

   Proof. Let A, B, and C are sets such that$B \subseteq C$.
   Let $(a,b) \in A \times B$. Then by the definition of $A \times B$, we know $a \in A$ and $b \in B$. Since $b \in B$ and $B \subset C$, it follows that $b \in C$. Thus, the (a,b) has the property that $a \in A$ and $b \in C$. Thus, it follows that $(a,b) \in A \times C$. □

4. If A, B, and C are sets, then $(A \cap B) - C = (A - C) \cap (B - C)$.

   Proof. Let A, B, and C be sets.

   First, we show that $(A \cap B) - C \subseteq (A - C) \cap (B - C)$.
   Let $x \in (A \cap B) - C$. Thus, $x \in A$ and $x \in B$ and $x \notin C$. Since $x \in A$ and $x \notin C$, it follows that $x \in (A - C)$. Similarly, since $x \in B$ and $x \notin C$, it follows that $x \in (B - C)$. Thus, $x \in (A - C)$ and $x \in (B - C)$, which implies $x \in (A - C) \cap (B - C)$.
   Since every element of $(A \cap B) - C$ is also an element of $(A - C) \cap (B - C)$, it follows that $(A \cap B) - C \subseteq (A - C) \cap (B - C)$.

   Second, we show that $(A - C) \cap (B - C) \subseteq (A \cap B) - C$.
   Let $x \in (A - C) \cap (B - C)$. Thus, $x \in (A - C)$ and $x \in (B - C)$. Since $x \in A - C$, we know $x \in A$. Since $x \in B - C$, we know $x \in B$. Since $x \in A$ and $x \in B$, we know $x \in A \cap B$.
   Further, since $x \in A - C$, we know $x \notin C$. So we have established that $x \in A \cup B$ and $x \notin C$. Thus, $x \in (A \cap B) - C$.
   We have shown that $x \in (A - C) \cap (B - C)$ implies $x \in (A \cap B) - C$. Thus, $(A - C) \cap (B - C) \subseteq (A \cap B) - C$.

   Since $(A \cap B) - C \subseteq (A - C) \cap (B - C)$ and $(A - C) \cap (B - C) \subseteq (A \cap B) - C$, we can conclude $(A - C) \cap (B - C) = (A \cap B) - C$. □